

Seeking Principled Diffeomorphic Hylogisms (3/6): Initialising Principled Constraints

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This version presents an updated specification for the pullback metric that only depends on the semantic warp. The appendices offer toy examples under the current and prior derivations. Should later scrutiny invalidate further claims, the author will continue to update this draft. The reader should not treat these early iterations as settled.

Abstract

This research note is the third in a series that examines the bridge between the physical universe's symmetric informational descriptions. En-route to a diffeomorphic connection from the state-configuration space to the semantic vector space, this note seeks a functorial hylogism, allowing the principled identification of physical structures across space and scales. The prior note discussed a diffeomorphism from the Borel σ -algebra proxy and state-configuration space, to the Borel-induced quotient coproduct. This note specifies the principles that constrain the mapping between the smooth quotient co-product proxy E_S^{reg} and the semantic vector space $\mathbb{V}$.

The $\mathcal{E}_S$ – $\mathbb{V}$ Distinction

The Intrinsic Space $\mathbb{V}_{int}$.

If $\mathcal{E}_S$ contains all physical forms, and if humans only ever observe physical forms, then little else can occupy a grounded semantic vector space. In fact, $\mathcal{E}_S$ may *be* the true SVS. Notice the type mismatch, however: $\mathcal{E}_S$ is not a vector space, but a family of such spaces - a vector bundle. Each B-region induces a vector space $Q_{S,B}$ whose axes originate from the bubbled coordinates $x \in B^1$.

Further, meaning and physical forms may be separable: one physical object may have two valid informational representations, an immaterial concept no physical instance. For example, ‘faith’ seems to have no physical form. This note holds, however, that for a given abstraction, a , there exists a collection of physical forms in $\mathcal{E}_{S,[a]}$ constituting a . This note rejects dual objects (sec.). Faith, for example, resides in form: in words, in thoughts, in prayers, etc.. So, while physical and informational objects may theoretically decouple, this note takes the dominant view in modern science that observable phenomena, including informational abstractions, are somehow physically present [1], [2], [3].²

To achieve that coupling while preserving the diffeomorphism, consider the necessary condition³:

$$\dim(\mathcal{E}_S) = \dim(\mathbb{V}). \quad (1)$$

¹See ap. A

²This position does not immediately imply the stronger stance that the physical universe *produces* the informational universe, only that the pair exist in pre-established harmony [4].

³Recall, a diffeomorphism is desirable for the initial $P \leftrightarrow I$ correspondence because (i) one recovers each Borel-configuration $[s]_{B_{\theta,c}}$ from its vector via the inverse h^{-1} , (ii) infinitesimal perturbations in the Borel-configuration induce infinitesimal semantic perturbations (permitting local analysis), and (iii) local differential properties may extend globally via continuous induction.

This constraint simplifies the hylogism. All semantically distinct objects reside in $\mathcal{E}_{\mathbb{S}}$ and their dimensions equate. Thus, all semantic objects are accessible by deforming (i) the focal field configuration, and (ii) the observation boundary. Consequently, $\dim(Q_{\mathbb{S},B})$ (local fibre) and $\dim(\mathcal{A}_{reg})$ (base space) combine in the final vector space. Locally, trivialise the vector bundle into a flat vector space via an orthogonal direct sum:

$$V_{intrinsic}|_U \cong U \times Q_{\mathbb{S},B_0} \subset \mathbb{R}^{\dim(\mathcal{A}_{reg})} \oplus Q_{S,B_0}. \quad (2)$$

Although contractibility⁴ of the regular base space $\mathcal{A}_{reg}$ would sufficiently extend this trivialisation globally, this note restricts its scope to the local triviality an observation window permits. This trivialisation implies each semantic vector $v \in \mathbb{V}_{int.}$ is composite with respect to the B-region parameters and field configurations. For approximate regular-cut and state parameters:

$$a_{reg,\theta} = [\theta_{\mathbf{B},1}, \dots, \theta_{\mathbf{B},|\Theta_{\mathbf{B}}|}, c] \quad (3)$$

$$\Theta_{\mathbb{S}} = [\theta_{\mathbb{S},1}, \dots, \theta_{\mathbb{S},|\Theta_{\mathbb{S}}|}] \quad (4)$$

Define a finite coordinate representation:

$$v = [a_{reg,\theta} ; \Theta_{\mathbb{S}}] \in \mathbb{V}_{int.} \quad (5)$$

If these claims hold, $\mathbb{V}_{int.}$ will have no ‘empty space’ to encode extra, non-physical semantics; $\mathcal{E}_{\mathbb{S}}$ is locally isomorphic to $\mathbb{V}_{int.}$ in the physical interaction’s immediate neighbourhood.

Extrinsic Spaces $\mathbb{V}_{ext.}$

If semantic differences are purely physical, the vector space of form-encodings $\mathbb{V}_{int.}$ is a valid semantic space. Yet, $\mathbb{V}_{int.}$ poses frictions (Table 1). For example, the label ‘chair’ becomes an equivalence class over different chair-like vectors $v_{chair,n} \in \mathbb{V}_{int.}$.

$$C_{chair} = \{v \in \mathbb{V}_{int.} : v \text{ represents a chair}\} \quad (6)$$

Operations on equivalence classes are more difficult than vector-wise operations, especially if class-members do not reside in connected regions; class equivalence need not preserve the vector-space structure. Semantics as humans know it becomes fragmented and mottled across smooth physical deformations. Further, $\mathbb{V}_{int.}$ does not directly express pluralistic semantic interpretations⁵, nor the brutal semantic differences that follow minute physical changes⁶.

Using $\mathbb{V}_{int.}$ as an SVS thus represents a distinctive philosophical stance. Any perceived variant $\mathbb{V}'_{int.}$ becomes an inexact distortion or a coarsening due to observation ‘error’⁷. In $\mathbb{V}_{int.}$, labels are unique elements; the *term* ‘chair’ is a coarse descriptor of all chair *objects*.

$$v_{chair}^{(label)} \neq v_{chair}^{(object)} \quad (7)$$

⁴A formal proof of global triviality for $\mathcal{E}_{\mathbb{S}}$ is deferred to later iterations.

⁵Though differing perceptions appear as physical observer-responses.

⁶For example, a chair whose legs each have an atomic-depth fracture on their cross section is no longer considered a chair - rather a health hazard.

⁷No calculation error necessarily exists. ‘Error’ here refers to the difference in meaning an observer ascribes relative to an object’s physical structure.

Complaint	Resolution
(1) $\mathbb{V}_{\text{int.}}$ separates semantically similar, but physically distinct objects (e.g. one chair moved to a different room.)	Physical form is meaning, declare a displaced chair a different object.
(2) $\mathbb{V}_{\text{int.}}$ does not group or class objects.	A group or class is a physical perceptive process in $\mathbb{V}_{\text{int.}}$.
(3) Minute physical changes in $\mathbb{V}_{\text{int.}}$ may cause brutal semantic shifts.	Physical form is meaning, slight physical deformations are exactly slight semantic differences.
(4) Multiple SVSs arise from different training corpora.	Any SVS not $\mathbb{V}_{\text{int.}}$ is a physical object in $\mathbb{V}_{\text{int.}}$.

Table 1: Complaints and resolutions disputing $\mathbb{V}_{\text{int.}}$ ’s validity.

Labels are thoughts, physical signs, or time-specific utterances. Two observers’ ‘chair’s therefore differ, each grounded in a unique physical context.⁸ These utterances are distinct in $P = \mathbb{R}^4$ and $\mathbb{V}_{\text{int.}}$ may fail to convey their semantic similarity. ‘So,’ asks the reader: ‘how might one group objects and labels?’ Perhaps the hylogism extends from $\mathbb{V}_{\text{int.}}$ to a separate $\mathbb{V}_{\text{ext.}}$. Formally, seek a map

$$h_{\text{ext}} : \mathbb{V}_{\text{int.}} \longrightarrow \mathbb{V}_{\text{ext.}} \quad (8)$$

A new space is unnecessary, however, because all conceivable forms reside in $\mathbb{V}_{\text{int.}}$. Instead, group like-vectors via metric deformations.⁹ Formalise this duality via the *pullback metric*. In the active view, the hylogism is a correspondence h into a target space with metric $g_{\mathbb{V}}$. In the passive view, the spaces are ontologically identical and one deforms a metric on $\mathbb{V}_{\text{int.}}$ by pulling back the metric $g_{\mathbb{V}}$ through the mapping derivative dh .

Mapping the physical space to a distinct semantic space with metric ($g_{\mathbb{V}}$) is equivalent to applying a localised metric deformation to $\mathbb{V}_{\text{int.}}$. This duality preserves the *semantic warping tensor* Ω and its *metric warp* $\tilde{\Omega}$:

$$h^* g_{\mathbb{V}} = \tilde{\Omega}_{\text{obs},p} \quad (9)$$

where:

- *Pullback Metric* - $h^* g_{\mathbb{V}}$: The geometric measure induced on $\mathbb{V}_{\text{int.}}$ by the sub-hylogism $h_{\text{ext.}}$ (abridged h). Equate the semantic target space projection with the localised physical deformation $dv_{\text{int.}}$.
- *Semantic Warp* - $\tilde{\Omega}$: A non-linear geometric perturbation. $\tilde{\Omega}$ scales the metric to compress or expand physical forms into semantic neighborhoods¹⁰, but only activates under an observer’s deformation.
- *Observer State* - $[s_{\text{obs}}]$: The transducing observer’s physical configuration. $[s_{\text{obs}}]$ parameterises $\tilde{\Omega}$, ensuring the metric deformation is observer-relative. For a null-response state (e.g., vacuum),

$$\tilde{\Omega}([s]_{\text{null-state}},p) = \mathbf{0}, \quad (10)$$

where null-responses arise from irregular inert objects (e.g. a total vacuum that does not respond differently to different physical objects.) These null responses lie beyond the diffeomorphic domain.

As no extraneous dimensions exist in a diffeomorphism, meaning cannot materialise when pushing a physical state into an unaccounted void.

⁸(E.g., when the author says ‘chair’ and the reader says ‘chair’ those are two different – albeit symmetric – symbols.)

⁹Irrespective of whether one instantiates a new space, or deforms the current space, the diffeomorphism requires these two geometric interpretations are mathematically identical.

¹⁰Note that high dimensionality is crucial for these disentanglements.

Questions of Meaning

Assuming Relative Causal Meaning

‘So,’ queries the reader, ‘where does this semantic warping tensor come from?’ To wrangle the intrinsic SVS into a semantic space, one must first ask ‘What *is* semantics?’ Typically, ‘semantics’ refers to the linguistic study of meaning – then ‘what *is* meaning?’ Philosophers have long studied meaning in a now labyrinthine literature. Traditional frameworks frequently bifurcate reality, examining meaning as (i) a referent to intangible concepts, (ii) a platonic ideal, or (iii) a subjective mental state differing from any physical basis [5][6][7]. These conventions often do not comply with the current purely geometric interpretation. For example, many are dualistic in the sense that they claim an object’s meaning can exist in the abstract, outside of the physical universe. Dual objects reside beyond the hylogism’s image (fig. 1).

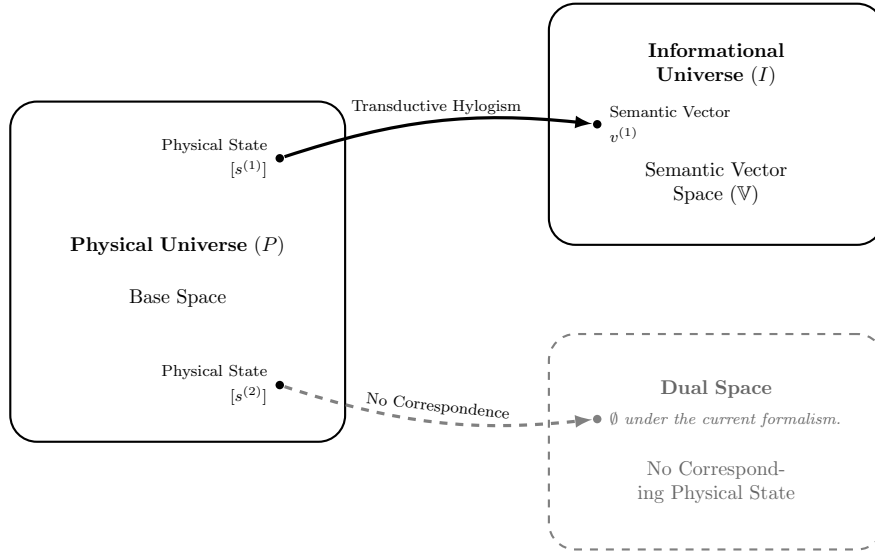


Figure 1: Comparison between physically grounded semantics and dualist semantics. The transduction pairs configurations with vectors. The dashed region illustrates semantic objects not generated by a physical configuration, lying beyond the coupled P, I universes.

As mentioned, this note rejects the existence of dual objects. While the current framework is agnostic to the it-bit relation’s *causal* direction, bijectivity is assumed. This stance aligns meaning with the geometric traditions in relational quantum mechanics [8] and teleosemantics [9]. Consider relative causal meaning:

Assumption 1 (Observer-Relative Causal Meaning). *An object o_1 has an observer-specific meaning relative to a witness o_2 , wholly conveyed by the structural change that o_1 enacts on o_2 .*

Assumption 1 is extreme. The implication is that any meaning one derives comes merely from physical interactions. A word obtains meaning when air oscillates the eardrum, a dream through synaptic changes from neuron-to-neuron, a chair when light reflects from surface-to-retina. Yet, the assumption is useful because the semantic warping tensor no longer needs a ‘divine’ external origin; the space of forms becomes its own interpretative lens.

Physical Impressions and Semantic Vectors

If meaning derives strictly from causal geometry, then each (perhaps every) physical interaction is a candidate vessel of semantic value. A drop of milk splashing into a cup of tea, the chaotic collision of plasma in a star’s core, and the tactile impact of a chair leg on a soft carpet all convey some physical perturbation that one could construe as the computation of meaning.

Still, to prevent this framework from collapsing into trivial panpsychism, one must mathematically distinguish between ‘dumb’ kinetic interactions and more structural cognition. One looks to the internal representations in artificial neural networks, and how those representations materialise as physical perturbations in the more recent spatial-only embedded architectures [10]. Consider the *asymmetric causal derivative* of two interacting Borel regions:

$$\delta s_{obs} = (\delta B_{obs}, \delta[s_{obs}]) \Big|_{B_{obs} \cap B_{exo} \neq \emptyset} \in T_{a_{obs}} \mathcal{A}_{reg} \oplus T_{[s_{obs}]} Q_{\mathbb{S}, B} \quad (11)$$

This composite derivative isolates the physical impression of an object on the observer:

- *(Composite) Causal Derivative* - δs_{obs} : The vector expressing the differential change in the observer’s structure. δs_{obs} is asymmetric, measuring only the impact on the observer, not the observed,
- *Variational Operator* - δ : As the base manifold P is a 4D spacetime, the regions B are static 4D world-tubes. Thus, δ evaluates counterfactual displacement (the geometric difference between an unperturbed trajectory and an interacted trajectory), not temporal progression,
- *Boundary Perturbation* - $\delta B_{obs} \in T_{a_{obs}} \mathcal{A}_{reg}$: The structural deviation of the observer’s world-tube threshold (e.g., a physical flinch, a shifting focal boundary, or mechanical degradation).
- *Field Perturbation* - $\delta[s_{obs}] \in T_{[s_{obs}]} Q_{\mathbb{S}, B}$: The localised change in the physical mass-energy configuration internal to the observer’s window (e.g., retinal photon absorption),
- *Overlap Condition* - $B_{obs} \cap B_{exo} \neq \emptyset$: The prerequisite local physical overlap. Without intersection, the continuous fields do not touch, and the causal derivative is null ($\delta s_{obs} = \mathbf{0}$).

When the base manifold is 4D spacetime, the derivative δs_{obs} does not trigger a time-varying trajectory. δs_{obs} encodes the interaction’s entire spatiotemporal cascade as a static object. Simply, this derivative captures the physical impression that a focal object leaves on a witness object over time. Meaning resides within.

The causal derivative helps to bridge from the space of forms to the semantic vector space. To see why, consider that two objects in $\mathcal{E}_{\mathbb{S}}$ interact to produce δs , so the physical impression is a binary operator:

$$\star : E_S \times E_S \longrightarrow TE_S, \quad (12)$$

$$((B_{exo}, [s]_{exo}), (B_{obs}, [s_{obs}])) \longmapsto (\delta B_{obs}, \delta[s_{obs}]) \in T_{(B_{obs}, [s_{obs}])} E_S. \quad (13)$$

Deriving Ω via Structural Jacobian Probing

Probing and Cognitive Responses

The semantic warping tensor Ω is the implicit constraint on each causal derivative’s realisation, given a focal observer. Explicitly derive Ω from the observer’s physical structure. Firstly, let the observer (e.g., an SE-renormalised neural network) be a bounded multi-field state,

$$e_{obs} = (B_{obs}, [s_{obs}]_{B_{obs}}) \in \mathcal{E}_{\mathbb{S}}. \quad (14)$$

The s-catalogue allows arbitrary observation windows and, during object interactions, one may require the boundary to enclose exactly that architecture. Otherwise, the region may contain non-cognitive field values (e.g., the air around a person's head).

$$\Phi_{obs} = F_{cog}([s_{obs}]) = F_{cog}([\phi_0]_{B_{obs}}, [\phi_1]_{B_{obs}}, \dots, [\phi_{N_\phi}]_{B_{obs}}). \quad (15)$$

Assume the continuous internal field Φ_{obs} inhabits a Hilbert space $\mathcal{H}$ (e.g., the vector-valued Lebesgue space $L^2(B_{obs}; \mathbb{R}^{N_\phi+1})$) equipped with the standard spatial inner product:

$$\langle \psi_1, \psi_2 \rangle_{\mathcal{H}} = \int_{B_{obs}} \psi_1(x) \cdot \psi_2(x) dx. \quad (16)$$

Because a local trivialisation represents the bundle patch in a flat vector space, its tangent space is locally isomorphic to that space. Thus, one represents the physical impression as a displacement vector in $\mathcal{H}$. Let $\Delta_{obs}(p) \in \mathcal{H}$ be that probe $p \in \mathcal{H}$ induced observer-relative response, both represented in the same trivialised space.

Formalise the response map and its Jacobian thus:

$$h_{obs} : \mathbb{V}_{int.} \longrightarrow \mathbb{V}_{ext.} \quad (17)$$

$$h_{obs}(p) = \Delta_{obs}(p) \quad (18)$$

$$J_{obs}(p) := D_p \Delta_{obs}(p) \quad (19)$$

$$dh_{obs}|_p = J_{obs}(p). \quad (20)$$

So,

$$dh_{obs}|_p : T_p \mathbb{V}_{int.} \longrightarrow T_{h_{obs}(p)} \mathbb{V}_{ext.} \quad (21)$$

maps a tangent probe-perturbations observer-response shifts. Use the standard pullback-metric construction from nonlinear elasticity. The operator

$$C_p := \left(dh_{obs}|_p \right)^* \left(dh_{obs}|_p \right)$$

is analogous to the right Cauchy–Green tensor. The pulled-back response geometry is the complete observer-relative geometry, define

$$\Omega_{obs}(p) := C_p.$$

Before evaluating the semantic warping tensor, one constrains Ω by principled necessity. A cognitive observer must maintain persistent internal representations despite environmental noise. For example, meaningless perturbations in the anoetic span demand that a cognitive architecture suppress these intrusions to preserve an internal semantic representation. Viewing a chair from different angles must not vastly change a person's cognitive activations; GDL invariance thus has a physical instantiation.

Asymptotic Warping Limits

An exact cancellation, $dh_{obs}|_p(u) = \mathbf{0}$ for some non-zero u , makes the mapping derivative non-injective, nullifying local invertibility and the keenly sought diffeomorphism [11]. The diffeomorphic construction therefore applies only on the regular region where $dh_{obs}|_p$ remains invertible. Simply, the observer's semantic warp must never reach zero between two distinctly observed object. In that case, the observer cannot distinguish between these objects and the diffeomorphism fails. For very similarly perceived objects, therefore, assume exact degenerate collapsing occurs only in an unreachable asymptotic limit. In this limit,

$$dh_{obs}|_p(u) = J_{obs}(p)u \approx \mathbf{0}. \quad (22)$$

Thus, morphing the probe's form along the anoetic direction produces little difference in the observer's response-geometry. The semantic warping tensor uses the operator's formal adjoint with respect to $\mathcal{H}$ to capture this dynamic:

$$\Omega_{obs}(p) := (dh_{obs}|_p)^* (dh_{obs}|_p) \quad (23)$$

$$\Omega_{obs}(p) = J_{obs}(p)^* J_{obs}(p) \quad (24)$$

$$\text{where} \quad \tilde{\Omega}_{obs,p}(u, w) := \langle u, \Omega_{obs}(p)w \rangle_{\mathcal{H}}. \quad (25)$$

Equip $\mathbb{V}_{ext.}$ with the response-space metric induced by the Hilbert inner product. The observer-conditioned local representative h_{obs} of h_{ext} then induces

$$(h_{obs}^* g_{ext.})_p(u, w) = g_{ext., h_{obs}(p)}(dh_{obs}|_p(u), dh_{obs}|_p(w)) \quad (26)$$

$$= \langle J_{obs}(p)u, J_{obs}(p)w \rangle_{\mathcal{H}} \quad (27)$$

$$= \langle u, \Omega_{obs}(p)w \rangle_{\mathcal{H}} \quad (28)$$

$$= \tilde{\Omega}_{obs,p}(u, w) \quad (29)$$

$$= g_{obs,p}(u, w). \quad (30)$$

Thus, the pullback metric is the complete observer-relative geometry:

$$h_{obs}^* g_{ext.} = g_{obs} = \tilde{\Omega}_{obs}. \quad (31)$$

Add no intrinsic metric to the pulled-back response geometry. Under asymptotic response conditions $dh_{obs}|_p(u) \approx \mathbf{0}$, the SWT approaches the zero response-geometry:

$$\Omega_{obs}(p)u = J_{obs}(p)^* J_{obs}(p)u \approx \mathbf{0}. \quad (32)$$

So,

$$g_{obs,p}(u, u) = \langle J_{obs}(p)u, J_{obs}(p)u \rangle_{\mathcal{H}} = \|J_{obs}(p)u\|_{\mathcal{H}}^2 \approx 0, \quad (33)$$

compressing $\mathbb{V}_{int.}$ axes¹¹ To measure the extent of the metric compression or expansion, consider the directional response ratio (relative to, say, the baseline geometry):

$$\rho_{obs,p}(u) := \frac{g_{obs,p}(u, u)}{\langle u, u \rangle_{\mathcal{H}}} = \frac{\|J_{obs}(p)u\|_{\mathcal{H}}^2}{\|u\|_{\mathcal{H}}^2}. \quad (34)$$

$0 < \rho_{obs,p}(u) < 1$ expresses compression, $\rho_{obs,p}(u) = 1$ preserves the baseline coordinate scale, and $\rho_{obs,p}(u) > 1$ expresses stretch.

Conclusion

This note documents an early attempt to convert the vector bundle $\mathcal{E}_{\mathbb{S}}$ into a warped semantic space to match a physical observer’s world view. Under this framework, semantic vectors arise as the physical impression a focal object leaves on an observer. One derives the warping tensor from an effective field over the truer fundamental reality¹². This note’s yield is a technical sketch of how the space of physical forms might come to occupy a valid vector space, and is pending further examination/validation. The next step in the series will discuss how abandoning the diffeomorphic restriction recovers coarser, more conventional SVS approximations. Specifically, spectral decomposition and eigenvalue ranks yield principled discretisations and coarsening thresholds.

Acknowledgments

AI models contributed to this work. While the author does not formally recognise AI co-authorship [12], [13], [14], [15], the author acknowledges the meaningful contributions of Gemini 3.1 Pro-Extended [16] and GPT 5.6 Sol-High [17]. Gemini-prompting has consistently yielded creative insights to the abstract reasoning process and as the primary AI-contributor, 3.1 Pro-Extended helped articulate these formalisms.

GPT 5.6 Sol-High initially suggested a Green-Lagrange (GL) strain construction for Ω ; the author later relinquished that analogy in the main response-only pullback derivation. 3.1 Pro-Extended and 5.6 Sol-High then helped replace the GL construction with the current right Cauchy-Green operator $J_{obs}^* J_{obs}$. Having reviewed these contributions independently, the author accepts full responsibility for the work and any remaining errors.

References

- [1] R. Landauer, “Irreversibility and heat generation in the computing process,” *IBM journal of research and development*, vol. 5, no. 3, pp. 183–191, 1961.
- [2] S. Lloyd, “Computational capacity of the universe,” *Physical Review Letters*, vol. 88, 2002. DOI: [10.1103/physrevlett.88.237901](https://doi.org/10.1103/physrevlett.88.237901).
- [3] J. A. Wheeler, “Information, physics, quantum: The search for links,” in *Complexity, Entropy, and the Physics of Information*, W. H. Zurek, Ed., Redwood City, CA: Addison-Wesley, 1990, pp. 3–28.
- [4] G. W. Leibniz, *The monadology and other philosophical writings.*, 1898.
- [5] G. Frege, “On sense and reference,” *The Philosophical Review*, vol. 57, no. 3, pp. 209–230, 1948, Originally published in German in 1892 as ‘Über Sinn und Bedeutung’.
- [6] Plato, *Republic*, trans. by G. M. A. Grube and C. D. C. Reeve. Indianapolis: Hackett Publishing Company, 1992, Book VII (The Allegory of the Cave) formalises the Theory of Forms.

¹¹The main innovation here is not needing to counteract the prior $g_{int.}$ metric in the g_{obs} pullback.

¹²(A simplification that the framework may later relax.)

- [7] D. J. Chalmers, “Facing up to the problem of consciousness,” *Journal of Consciousness Studies*, vol. 2, no. 3, pp. 200–219, 1995.
- [8] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics*, vol. 35, no. 8, pp. 1637–1678, 1996.
- [9] R. G. Millikan, “Biosemantics,” *The Journal of Philosophy*, vol. 86, no. 6, pp. 281–297, 1989.
- [10] P. Tohouri. “Computationally constrained renormalisation in virtual brain reconstruction.” version 11. figshare preprint. [Online]. Available: <https://doi.org/10.6084/m9.figshare.31764994>.
- [11] J. M. Lee, *Introduction to Smooth Manifolds* (Graduate Texts in Mathematics). New York, NY: Springer New York, 2013, vol. 218. DOI: [10.1007/978-1-4419-9982-5](https://doi.org/10.1007/978-1-4419-9982-5).
- [12] N. Levy, “Responsibility is not required for authorship,” *Journal of Medical Ethics*, jme-2024–109 912, 2024. DOI: [10.1136/jme-2024-109912](https://doi.org/10.1136/jme-2024-109912).
- [13] G. E. Kaebnick, D. C. Magnus, A. Kao, et al., “Editors’ statement on the responsible use of generative ai technologies in scholarly journal publishing,” *Medicine, Health Care and Philosophy*, vol. 26, pp. 499–503, 2023. DOI: [10.1007/s11019-023-10176-6](https://doi.org/10.1007/s11019-023-10176-6).
- [14] J. Y. Lee, “Can an artificial intelligence chatbot be the author of a scholarly article?” *Science Editing*, vol. 10, pp. 7–12, 2023. DOI: [10.6087/kcse.292](https://doi.org/10.6087/kcse.292).
- [15] B. D. Lund and K. T. Naheem, “Can chatgpt be an author? a study of artificial intelligence authorship policies in top academic journals,” *Learned Publishing*, vol. 37, pp. 13–21, 2023. DOI: [10.1002/leap.1582](https://doi.org/10.1002/leap.1582).
- [16] Google DeepMind, “Gemini 3.1 pro model card,” Google DeepMind, Tech. Rep., Feb. 2026. [Online]. Available: <https://deepmind.google/models/model-cards/gemini-3-1-pro/>.
- [17] OpenAI, “Gpt-5.6 system card,” OpenAI, Tech. Rep., 2026. [Online]. Available: <https://deploymentsafety.openai.com/gpt-5-6/gpt-5-6.pdf>.

Appendices

A Space Disambiguation

Term	Type	Element	Description
Physical Reality (P)	$\mathbb{R}^D$	$x \in P$	A smooth base manifold ($\mathbb{R}^4$ for spacetime) on which all fields reside.
Borel σ -algebra ($\mathcal{B}(P)$)	Measurable Space	$B \subset P$	The collection of measurable subsets on P (e.g., world-tubes) that distinguish objects.
$\mathcal{E}$ Parameterisation Space ($\mathcal{A}_{reg}$)	$\mathbb{R}^{dim(\mathcal{A}_{reg})}$	$(\theta, c) \in \mathcal{A}_{reg}$	The continuous domain that induces smooth, regular B -regions via scalar fields and smooth c -cuts.
State Configuration Space ($\Gamma(P, \mathbb{S})$)	Function Space (∞ -dim.)	$s \in \Gamma(P, \mathbb{S})$	An infinite-dimensional, perfectly flat function space comprising the unconstrained global catalogue of all possible field evaluations.
Borel-Induced Space ($\Gamma(P, \mathbb{S})/K_B$)	Quotient	Quotient Space (∞ -dim.)	$[s]_B = s + K_B$ A flat quotient vector space where external universe variations collapse into the linear zero-subspace K_B .
Quotient Cartesian Product ($\prod_{n_\phi} \mathcal{Q}_{n_\phi, B}$)	Vector Space (∞ -dim.)	$([\rho]_B, [\phi_1]_B, \dots)$	A flat vector space stitching independent fundamental fields together, continuous across physical components.
The s -Catalogue ($\mathcal{E}_{\mathbb{S}}$)	Smooth Vector Bundle	$(B, [s]_B)$	A smooth vector bundle utilising $\mathcal{A}_{reg}$ as its base and the flat quotient spaces as its fibers. Synonymous with $\mathbb{V}_{int.}$.
Int./Ext. Semantic Vector Space ($\mathbb{V}$)	Vector Space (∞ -dim.)	$v \in \mathbb{V}$	A flat vector space trivialised from the $\mathcal{E}_{\mathbb{S}}$ bundle via an orthogonal direct sum. $\mathbb{V}_{int.}$ is the structural base, $\mathbb{V}_{ext}$ its observer-relative metric deformation via Ω .

Table 2: Space-Type Taxonomy

B Simplified Examples

B.1 Exogenous Observer, Perturbed Two-Field Configuration.

To ground the theory with a worked example, consider a minimal hylogical pipeline mapping a physical field to the semantic space. The physical space P is a 1D coordinate line ($x \in \mathbb{R}$). Assume a total state configuration $s \in \Gamma(P, \mathbb{S})$ comprising two fundamental scalar fields: the mass-density $\phi_0(x)$ and an environmental noise field $\phi_1(x)$ (e.g., macroscopically-observed temperature). Define a uniform global base state for a focal object:

$$\phi_0(x) = 1, \quad \phi_1(x) = 0 \quad \forall x \in \mathbb{R}. \quad (35)$$

The probe exists in a bounded region given by the Borel set $B_p = [0, 1]$. Restricting the fields to this window yields $\phi_0|_{B_p}$ and $\phi_1|_{B_p}$. Applying the quotient map collapses all external universal variations into the equivalence class $[s]_{B_p}$. Under the chosen local trivialisation, this equivalence class is represented in the intrinsic Hilbert space $\mathcal{H} = L^2(B_p; \mathbb{R}^2)$, equipped with a spatial inner product. The base point $p \in \mathcal{H}$ is the unperturbed state within the window:

$$p(x) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{for } x \in [0, 1]. \quad (36)$$

Perturb p by $u \in \mathcal{H}$. Let u contain a macroscopic variation $u_0(x) = u_{\phi_0}$ and a noise perturbation $u_1(x) = u_{\phi_1}$:

$$u(x) = \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} \implies p(x) + u(x) = \begin{bmatrix} 1 + u_{\phi_0} \\ 0 + u_{\phi_1} \end{bmatrix}. \quad (37)$$

The probe and its perturbed variant both interact with the observer's cognitive field. For simplicity, the observer interaction is assumed exogenous; this example does not yet instantiate an observer internally. Assume that when the two probes p and $p + u$ impact the observer, the structural Jacobian is:

$$J_{obs}(p) = \begin{bmatrix} \frac{\partial \Delta_0}{\partial p_0} & \frac{\partial \Delta_0}{\partial p_1} \\ \frac{\partial \Delta_1}{\partial p_0} & \frac{\partial \Delta_1}{\partial p_1} \end{bmatrix}. \quad (38)$$

For this simplified example, assume h_{obs} is affine along the segment joining p and $p + u$. The Jacobian is thus constant along this segment, so the finite difference between the two observer-response geometries satisfies

$$h_{obs}(p + u) - h_{obs}(p) = J_{obs}(p)u. \quad (39)$$

Apply the mapping derivative $dh_{obs}|_p(u) = J_{obs}(p)u$ to the perturbation:

$$dh_{obs}|_p(u) = J_{obs}(p) \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} \quad (40)$$

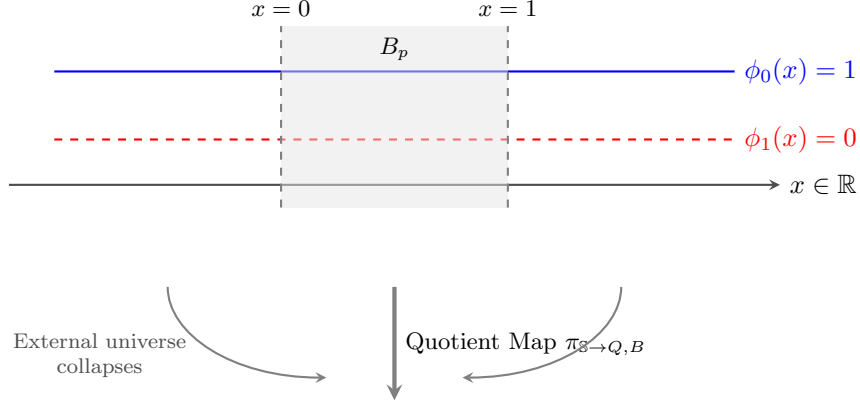
$$= \begin{bmatrix} \frac{\partial \Delta_0}{\partial p_0} \cdot u_{\phi_0} + \frac{\partial \Delta_0}{\partial p_1} \cdot u_{\phi_1} \\ \frac{\partial \Delta_1}{\partial p_0} \cdot u_{\phi_0} + \frac{\partial \Delta_1}{\partial p_1} \cdot u_{\phi_1} \end{bmatrix}. \quad (41)$$

Now, assume the observer's Jacobian conveys that a probe's mass perturbation affects only the final mass-response and likewise that temperature perturbations affect only the final temperature-response:

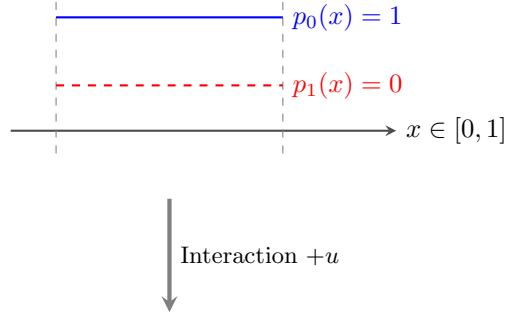
$$J_{obs}(p) = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \quad |a| > 1, \quad 0 < |b| < 1. \quad (42)$$

Whereas the mass response amplifies, the temperature response compresses, thereby suppressing “meaningless” thermal changes such that $J_{obs}(p)u_{anoetic} \approx \mathbf{0}$. Arbitrarily setting $a = 2$ and $b = \epsilon$, where $0 < \epsilon \ll 1$, yields:

1. Global Field Configuration $s \in \Gamma(P, \mathbb{S})$



2. Base Point $p \in \mathcal{H} = L^2(B_p; \mathbb{R}^2)$



3. Perturbed State $(p + u) \in \mathcal{H}$

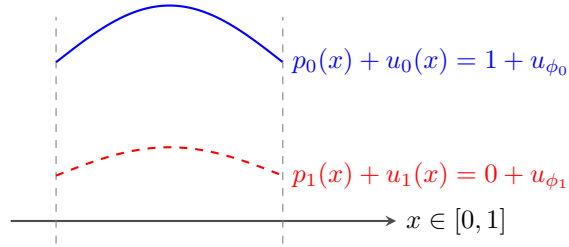


Figure 2: Simple worked example with state field quotienting and two-field perturbation. Top: macroscopic mass and temperature exist as a field configuration on $\mathbb{R}$; middle: the two fields are quotiented into a probe object defined by a trivialised equivalence class on the interval $[0, 1]$; bottom: a perturbation u distorts the probe, possibly unevenly across fields.

$$dh_{obs}|_p(u) = \begin{bmatrix} 2u_{\phi_0} \\ \epsilon u_{\phi_1} \end{bmatrix}. \quad (43)$$

In words, a perturbation along the probe's mass axis produces a response-geometry difference twice as large. Along the temperature axis, the observer's response-geometry is almost invariant under the probe perturbation. The observer therefore has a near-fixed reaction despite temperature differences. Term the first response “noetic” due to the observer's keen response, and the second “anoetic” due to the perturbation's near-imperceptibility.

To examine asymptotic semantic compression, consider the probe-shift

$$u(x) = \begin{bmatrix} 0 \\ 10 \end{bmatrix}. \quad (44)$$

The corresponding observer-response geometric difference is

$$dh_{obs}|_p(u) = J_{obs}(p)u = \begin{bmatrix} 0 \\ 10\epsilon \end{bmatrix} \approx \mathbf{0}. \quad (45)$$

The semantic warping tensor is the complete pullback-metric operator:

$$\Omega_{obs}(p) := (dh_{obs}|_p)^* (dh_{obs}|_p) \quad (46)$$

$$= J_{obs}(p)^* J_{obs}(p) \quad (47)$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & \epsilon^2 \end{bmatrix}. \quad (48)$$

Applying $\Omega_{obs}(p)$ to the anoetic probe direction yields

$$\Omega_{obs}(p)u = \begin{bmatrix} 0 \\ 10\epsilon^2 \end{bmatrix} \approx \mathbf{0}. \quad (49)$$

Evaluating the magnitude (squared length) of this perturbation under the observer-relative metric gives:

$$g_{obs,p}(u, u) = \langle J_{obs}(p)u, J_{obs}(p)u \rangle_{\mathcal{H}} \quad (50)$$

$$= \|J_{obs}(p)u\|_{\mathcal{H}}^2 \quad (51)$$

$$= 100\epsilon^2 \approx 0. \quad (52)$$

The same operator exhibits stretch along the mass axis and compression along the temperature axis. For the unit coordinate perturbations $e_0 = [1 \ 0]^T$ and $e_1 = [0 \ 1]^T$:

$$g_{obs,p}(e_0, e_0) = 4, \quad (53)$$

$$g_{obs,p}(e_1, e_1) = \epsilon^2. \quad (54)$$

Thus, the observer stretches mass-response lengths by a factor of 2 and compresses temperature-response lengths by a factor of ϵ . The two probes whose temperature components differ by 10 are separated by observer-relative distance 10ϵ , with squared distance $100\epsilon^2$. They asymptotically collocate $\epsilon \rightarrow 0$. At $\epsilon = 0$, exact collapse occurs, the pullback metric is degenerate along the temperature axis, and the local diffeomorphism fails.

B.2 Minimal Dynamical Observer Example

The preceding example assumes a structural Jacobian J_{obs} . Here, derive that Jacobian from a minimal observer response law. Retain the two-field probe representation on $B_p = [0, 1]$, with

$$H = L^2(B_p; \mathbb{R}^2). \quad (55)$$

For simplicity, restrict the example to the constant-function subspace

$$H_c := \{q \in H : q(x) = \bar{q} \text{ for some } \bar{q} \in \mathbb{R}^2\} \cong \mathbb{R}^2. \quad (56)$$

Since $|B_p| = 1$, the L^2 inner product on H_c coincides with the standard Euclidean inner product on its coordinate representation. Write a probe as

$$q = \begin{bmatrix} q_0 \\ q_1 \end{bmatrix}, \quad (57)$$

where q_0 denotes the mass-density component and q_1 the environmental-noise component. Let the observer have two reduced internal response coordinates

$$y(t) = \begin{bmatrix} y_0(t) \\ y_1(t) \end{bmatrix} \in \mathbb{R}^2. \quad (58)$$

While the probe q is presented, assume the observer obeys the stable linear response law

$$\dot{y}(t) = -\zeta y(t) + Kq, \quad (59)$$

where

$$\zeta = \begin{bmatrix} \lambda_0 & 0 \\ 0 & \lambda_1 \end{bmatrix}, \quad K = \begin{bmatrix} k_0 & 0 \\ 0 & k_1 \end{bmatrix}, \quad \lambda_0, \lambda_1 > 0. \quad (60)$$

Since the eigenvalues of $-\zeta$ are strictly negative, the observer converges to the unique stable equilibrium satisfying

$$0 = -\zeta y_\infty(q) + Kq. \quad (61)$$

Hence,

$$y_\infty(q) = \zeta^{-1} Kq \quad (62)$$

$$= \begin{bmatrix} k_0/\lambda_0 & 0 \\ 0 & k_1/\lambda_1 \end{bmatrix} q. \quad (63)$$

Define the observer-response map by this settled response:

$$h_{\text{obs}}(q) := y_{\infty}(q). \quad (64)$$

The structural Jacobian then follows from the response law:

$$J_{\text{obs}}(q) := Dh_{\text{obs}}(q) \quad (65)$$

$$= \zeta^{-1}K \quad (66)$$

$$= \begin{bmatrix} k_0/\lambda_0 & 0 \\ 0 & k_1/\lambda_1 \end{bmatrix}. \quad (67)$$

Now assume that the observer responds strongly to mass perturbations and weakly to environmental noise, with gain-to-decay ratios

$$\frac{k_0}{\lambda_0} = 2, \quad \frac{k_1}{\lambda_1} = \varepsilon, \quad 0 < \varepsilon \ll 1. \quad (68)$$

Then,

$$J_{\text{obs}} = \begin{bmatrix} 2 & 0 \\ 0 & \varepsilon \end{bmatrix}. \quad (69)$$

For the base probe

$$p = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (70)$$

and perturbation

$$u = \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix}, \quad (71)$$

the settled observer-response difference is

$$h_{\text{obs}}(p + u) - h_{\text{obs}}(p) = J_{\text{obs}}u \quad (72)$$

$$= \begin{bmatrix} 2u_{\phi_0} \\ \varepsilon u_{\phi_1} \end{bmatrix}. \quad (73)$$

Equip the reduced observer-response space with its standard Euclidean metric. Pulling this geometry back through h_{obs} yields

$$g_{\text{obs},p}(u, w) = \langle J_{\text{obs}}u, J_{\text{obs}}w \rangle. \quad (74)$$

The associated semantic-warp operator is therefore

$$\Omega_{\text{obs}} := J_{\text{obs}}^* J_{\text{obs}} \quad (75)$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & \varepsilon^2 \end{bmatrix}. \quad (76)$$

For the coordinate directions

$$e_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (77)$$

one obtains

$$g_{\text{obs}}(e_0, e_0) = 4, \quad (78)$$

$$g_{\text{obs}}(e_1, e_1) = \varepsilon^2. \quad (79)$$

So, equally sized perturbations along the two probe coordinates receive different observer-relative lengths. The distinction is no longer introduced directly at the metric stage: it follows from the observer's settled response law. High gain relative to decay stretches the first probe direction; low gain relative to decay compresses the second. For every finite $\varepsilon > 0$,

$$\det J_{\text{obs}} = 2\varepsilon > 0, \quad (80)$$

so h_{obs} remains locally invertible on this reduced two-dimensional example. In the asymptotic limit

$$\varepsilon \rightarrow 0, \quad (81)$$

the observer becomes insensitive to the second probe direction:

$$J_{\text{obs}} e_1 \rightarrow 0, \quad (82)$$

$$g_{\text{obs}}(e_1, e_1) \rightarrow 0. \quad (83)$$

This recovers the anoetic compression described in the main construction. The local chain is now:

$$v_{\text{obs}} \text{ (dynamics)} \longrightarrow h_{\text{obs}} \longrightarrow Dh_{\text{obs}} \longrightarrow J_{\text{obs}}^* J_{\text{obs}} \longrightarrow g_{\text{obs}}. \quad (84)$$

The example does not independently derive which physical directions an observer ought to amplify or suppress: those sensitivities remain assumptions of the chosen response dynamics. It does, however, derive the structural Jacobian and consequent metric warp from an explicit observer law rather than imposing the Jacobian separately.

Supplementary Materials

C Ensuring $\mathbb{V}_{int}$ Structure Preservation

*This section contains a prior derivation for the pullback metric h^*g_V . The derivation arises under the assumption that the observer's metric g_{obs} sums the intrinsic metric with the semantic warp $g_{int} + \tilde{\Omega}_{obs,p}$. Degeneracy in $\tilde{\Omega}_{obs,p}$ thus does not collapse $\mathbb{V}_{int}$; the degeneracy condition is instead $J_{obs}(p)u = -u$. So, for the observer to collapse two like-forms and spoil the diffeomorphism under this specification, their internal response must compensate for the probe-deformation.*

Contrariwise, the main body's Δ -only derivation relies on a more direct non-degeneracy assumption: namely that an object must respond differently to different forms¹³. Given g_{int} no longer preserves baseline topology, probes not interacting with the observer may then collapse into indistinguishability such that only a local diffeomorphism persists.

These modeling choices turn on taste. The Δ -only method is calculable with only observer-relative quantities. So, despite permitting degenerate non-interacting cases, the Δ -only derivation aligns more closely with observer-relative causal meaning (assumption 1), than the form shown below.

Formalise the deformation and the Jacobian thus:

$$h_{obs}(p) = p + \Delta_{obs}(p) \quad (85)$$

$$J_{obs}(p) := D_p \Delta_{obs}(p) \quad (86)$$

$$dh_{obs}|_p = I + J_{obs}(p). \quad (87)$$

The cancellation $dh_{obs}|_p(u) = \mathbf{0}$ remains a (macroscopic) asymptote.

$$dh_{obs}|_p(u) = u + J_{obs}(p)u \approx \mathbf{0} \quad (88)$$

$$J_{obs}(p)u \approx -u \quad (89)$$

The semantic warping tensor still uses the operator's formal adjoint wrt. $\mathcal{H}$:

$$\Omega_{obs}(p) := (dh_{obs}|_p)^*(dh_{obs}|_p) - I \quad (90)$$

$$\Omega_{obs}(p) = (J_{obs}(p) + J_{obs}(p)^*) + J_{obs}(p)^* J_{obs}(p) \quad (91)$$

$$\text{where } \tilde{\Omega}_{obs,p}(u, w) := \langle u, \Omega_{obs}(p)w \rangle_{\mathcal{H}}. \quad (92)$$

Identify the intrinsic metric, again with the Hilbert inner product,

$$g_{int.,p}(u, w) := \langle u, w \rangle_{\mathcal{H}}, \quad (93)$$

¹³A restrictive assumption that the author intends to relax later.

the observer-conditioned local representative h_{obs} of h_{ext} induces

$$(h_{obs}^* g_V)_p = g_{obs,p} = g_{int.,p} + \tilde{\Omega}_{obs,p}. \quad (94)$$

Under homeostatic asymptotic compensation $dh_{obs}|_p(u) \approx \mathbf{0}$, the SWT applied to the anoetic vector returns the negative shift:

$$\Omega_{obs}(p)u \approx -u \quad (95)$$

The observer-relative metric is:

$$g_{obs,p}(u, w) = \langle u, [I + \Omega_{obs}(p)] w \rangle_{\mathcal{H}} \quad (96)$$

By substituting $\Omega_{obs}(p)u \approx -u$, the metric value is $g_{obs,p}(u, u) \approx \langle u, u - u \rangle_{\mathcal{H}} = 0$, compressing relevant $\mathbb{V}_{int.}$ axes. Notice, however, that in this asymptotic, degenerate case

$$g_{obs,p}(u, u) \approx 0, \quad (97)$$

So, topology collapses due to the observer's homeostatic cancellation of probe differences. The brain cancels perceived changes, but the underlying physical geometry still reflects $\mathbb{V}_{int.}$

D Simplified Examples

This section follows the worked examples in appendix B with the alternate metric from section C.

D.1 Exogenous Observer, Perturbed Two-Field Configuration.

To repeat the prior worked example, set the fields:

$$\phi_0(x) = 1, \quad \phi_1(x) = 0 \quad \forall x \in \mathbb{R} \quad (98)$$

The base probe $p \in \mathcal{H}$ is the unperturbed state within the window:

$$p(x) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{for } x \in [0, 1] \quad (99)$$

Perturb p by $u \in \mathcal{H}$. Let u contain a macroscopic variation $u_0(x) = u_{\phi_0}$ and noise perturbation $u_1(x) = u_{\phi_1}$:

$$u(x) = \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} \implies p(x) + u(x) = \begin{bmatrix} 1 + u_{\phi_0} \\ 0 + u_{\phi_1} \end{bmatrix} \quad (100)$$

The structural Jacobian is:

$$J_{obs}(p) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \left(J_{obs}(p) = \begin{bmatrix} \frac{\partial \Delta_0}{\partial p_0} & \frac{\partial \Delta_0}{\partial p_1} \\ \frac{\partial \Delta_1}{\partial p_0} & \frac{\partial \Delta_1}{\partial p_1} \end{bmatrix} \right) \quad (101)$$

Apply the mapping derivative $dh_{obs}|_p(u) = u + J_{obs}(p)u$ to the perturbation:

$$dh_{obs}|_p(u) = \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} \quad (102)$$

$$dh_{obs}|_p(u) = \begin{bmatrix} u_{\phi_0} \\ u_{\phi_1} \end{bmatrix} + \begin{bmatrix} u_{\phi_0} \\ -u_{\phi_1} \end{bmatrix} = \begin{bmatrix} 2u_{\phi_0} \\ 0 \end{bmatrix} \quad (103)$$

To verify semantic collapse, again consider an asymmetric (trivially-chosen) probe-shift $u(x) = [0 \ 10]^T$. The mapping derivative yields $dh_{obs}|_p(u) = \mathbf{0}$. The semantic warping tensor $\Omega_{obs}(p) := (dh_{obs}|_p)^*(dh_{obs}|_p) - I$ applied to u is:

$$\Omega_{obs}(p)u = (dh_{obs}|_p)^*(\mathbf{0}) - u = -u \quad (104)$$

Evaluating the magnitude (squared length) of this perturbation under the observer's metric yields:

$$g_{obs,p}(u, u) = \langle u, u + \Omega_{obs}(p)u \rangle_{\mathcal{H}} = \langle u, u - u \rangle_{\mathcal{H}} = 0 \quad (105)$$

This result demonstrates a homeostatic limit that collapses the baseline v_{int} geometry. The observer cognitively cancels the probe-shift in this new limit.